

Normal Acid-Base Balance

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DEFINITION

The acid-base status of the body is carefully regulated to maintain arterial pH between 7.35 and 7.45 and intracellular pH between 7.0 and 7.3. This regulation occurs in the setting of continuous production of acidic metabolites and is accomplished by intracellular and extracellular buffering processes with respiratory and renal regulatory mechanisms. This chapter reviews the normal physiology of acid-base homeostasis.

NET ACID PRODUCTION

Both acid and alkali are generated from diet. Lipid and carbohydrate metabolism results in production of carbon dioxide (CO_2), a volatile acid, at the rate of approximately 15,000 mmol/day. Protein metabolism yields amino acids, which can be metabolized to form nonvolatile acid and alkali. Amino acids such as lysine and arginine yield acid on metabolism, whereas the amino acids glutamate and aspartate and organic anions such as acetate and citrate generate alkali. Sulfur-containing amino acids (methionine, cysteine) are metabolized to sulfuric acid (H_2SO_4), and organophosphates are metabolized to phosphoric acid (H_3PO_4). In general, animal foods are high in proteins and organophosphates and provide a net acid diet; plant foods are higher in organic anions and provide a net alkaline load. In addition to acid and alkali generated from diet, there is a small daily production of organic acids, including acetic acid, lactic acid, and pyruvic acid. Also, a small amount of acid is generated by the excretion of alkali into the stool. Under normal circumstances, daily net nonvolatile acid production is approximately 1 mmol of hydrogen ions (H^+) per kilogram of body weight (Fig. 12.1).

BUFFER SYSTEMS IN REGULATION OF PH

Intracellular and extracellular buffer systems minimize the change in pH during the addition of acid or base equivalents but do not remove acid or alkali from the body. The most important buffer system is bicarbonate ion and carbon dioxide (HCO_3^- - CO_2). In this system, carbon dioxide concentration $[\text{CO}_2]$ is maintained at a constant level set by respiratory control. Addition of acid (HA) leads to conversion of HCO_3^- to CO_2 according to the reaction $\text{HA} + \text{NaHCO}_3 \rightarrow \text{NaA} + \text{H}_2\text{O} + \text{CO}_2$. HCO_3^- is consumed, but $[\text{CO}_2]$ does not change because this is maintained by respiration. The net result is that the acid load has been buffered and pH changes are minimal.

Although the HCO_3^- - CO_2 buffer system is the most important of the buffers in extracellular fluid (ECF), other buffers such as plasma proteins and phosphate ions also participate in the maintenance of a stable pH. During metabolic acidosis, the skeleton becomes a major

buffer source as acid-induced dissolution of bone apatite releases alkaline Ca^{2+} salts and HCO_3^- into the ECF. With chronic metabolic acidosis, this can result in osteomalacia and osteoporosis. The calcium released can result in hypercalciuria and an increased likelihood of renal stones. Within the intracellular fluid (ICF) compartment, pH is maintained by intracellular buffers such as hemoglobin, cellular proteins, organophosphate complexes, and HCO_3^- as well as by the $\text{H}^+ - \text{HCO}_3^-$ mechanisms that transport acid and alkali in and out of the cell.

RESPIRATORY SYSTEM IN REGULATION OF PH

Removal of acid or alkali from the body is accomplished by the lungs and kidneys. The lungs regulate CO_2 tension (Pco_2), and the kidneys regulate serum bicarbonate concentration $[\text{HCO}_3^-]$. Although the $[\text{HCO}_3^-]$ - CO_2 buffer system is not the only buffer system, all extracellular buffer systems are in equilibrium. Because serum $[\text{HCO}_3^-]$ is much greater than that of other buffers, changes in the HCO_3^- - CO_2 buffer pair easily titrate other buffer systems and thus set pH. The Henderson-Hasselbalch equation explains how the lungs and kidneys function in concert: pH is determined by the ratio of HCO_3^- to CO_2 . Conditions associated with similar fractional changes in $[\text{HCO}_3^-]$ and $[\text{CO}_2]$, such as when both are halved, will not change blood pH.

The lungs defend pH by altering alveolar ventilation, which alters the CO_2 excretion rate and thereby controls the arterial CO_2 tension (Paco_2) of body fluids. Systemic acidosis stimulates the respiratory center, resulting in increased respiratory drive that lowers the Paco_2 . As a result, the fall in blood pH is less than would have occurred in the absence of respiratory compensation. If the fractional change in Pco_2 were similar to that in serum $[\text{HCO}_3^-]$, blood pH would not change. However, respiratory compensation rarely normalizes blood pH, and thus the fractional change in Pco_2 is less than the change in serum $[\text{HCO}_3^-]$. Quantitatively, the normal respiratory response in metabolic acidosis is a 1.2 mm Hg decrease in Paco_2 for every 1 mmol/L decrease in HCO_3^- ; the increase in Paco_2 in response to metabolic alkalosis averages 0.7 mm Hg for every 1 mmol/L increase in HCO_3^- above baseline.¹

KIDNEY REGULATION OF pH

Buffer systems and respiratory excretion of CO_2 help maintain normal acid-base balance, but the kidneys provide a critical role in acid-base homeostasis. The kidneys normally generate sufficient net acid excretion (NAE) to balance nonvolatile acid produced from normal metabolism. NAE has three components, titratable acids, ammonium (NH_4^+), and bicarbonate, and is calculated by the following formula:

$$\text{NAE} = V (\text{U}_{\text{Am}} + \text{U}_{\text{TA}} - \text{U}_{\text{HCO}_3^-})$$

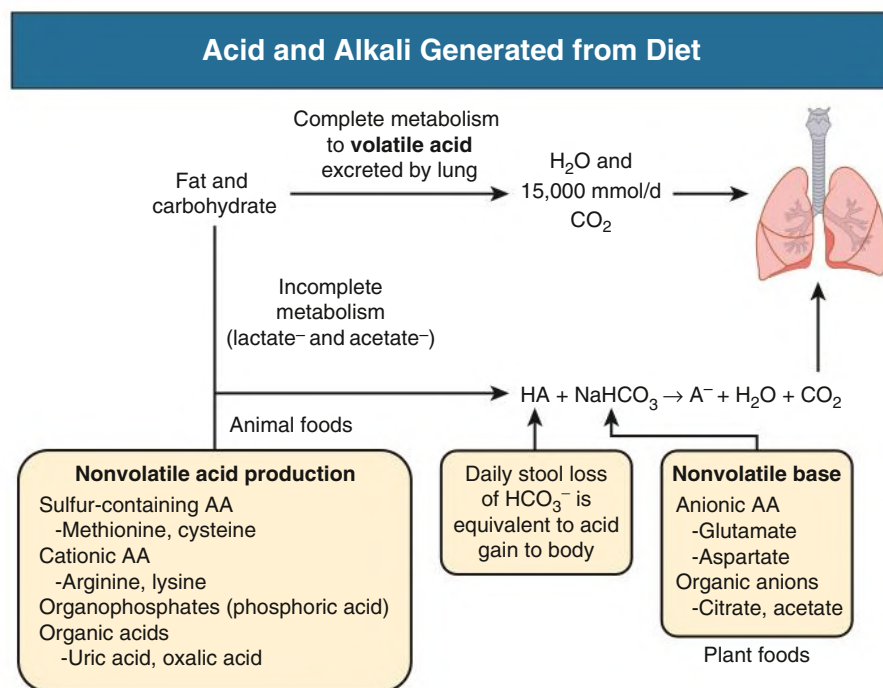


Fig. 12.1 Acid and Alkali Generated From the Diet. A nonvolatile acid is an acid produced from sources other than CO_2 and is not excreted by the lungs. Nonvolatile acids are produced from incomplete metabolism of carbohydrates, fats, and proteins and from metabolism of animal foods. Plant foods tend to produce an alkali load. AA, Amino acids.

where $U_{\text{Am}}V$ is the rate of NH_4^+ excretion, $U_{\text{TA}}V$ is the rate of titratable acid excretion, and $U_{\text{HCO}_3^-}V$ is the rate of HCO_3^- excretion. Under basal conditions, approximately 40% of NAE is in the form of titratable acids and 60% is in the form of ammonia (NH_3); urinary bicarbonate concentrations and excretion are essentially zero under normal conditions.

Titrateable acidity refers to weak acids filtered at the glomerulus that can act as buffers in the urine. These buffers are referred to as titrateable because they are measured by determining the amount of alkali required to titrate the urine back to a pH of 7.4. To serve as a titrateable buffer, a buffer must have a pK_a near the range of tubular fluid pH. The most important titrateable buffer is phosphate ($\text{HPO}_4^{2-} \leftrightarrow \text{H}_2\text{PO}_4^-$) because it has a favorable pK_a of 6.80 and there is a relatively high rate of urinary excretion. However, when acid production increases, the increase in acid excretion is almost entirely caused by an increase in excretion of NH_4^+ because the ability to increase urinary phosphate is limited.

KIDNEY TRANSPORT MECHANISMS OF HYDROGEN AND BICARBONATE IONS

Glomerulus

The glomerulus is not normally considered a participant in acid-base regulation. However, the glomerulus filters an amount of HCO_3^- equivalent to serum $[\text{HCO}_3^-]$ multiplied by the glomerular filtration rate (GFR). Under normal circumstances, the filtered load of HCO_3^- averages approximately 4000 mmol/day. Normal acid-base homeostasis requires both the reabsorption of this filtered bicarbonate and the generation of "new" bicarbonate; the latter replenishes bicarbonate and other alkaline buffers consumed in the process of titrating endogenous acid production. From the standpoint of prevention or correction of acidosis, GFR is not regulated by alterations in acid or base and therefore does not contribute to acid-base homeostasis.

Proximal Tubule

The proximal tubule reabsorbs approximately 80% of the filtered load of HCO_3^- . In addition, by titration of luminal pH from 7.4 down to approximately 6.7, the majority of phosphate, the major form of titrateable acid, is titrated to its acid form. Finally, ammonia synthesis occurs in the proximal tubule.

Fig. 12.2 shows the acid-base transport mechanisms of the proximal tubule cell. HCO_3^- absorption from the tubular lumen is mediated by H^+ secretion across the membrane.² This H^+ secretion is active in that the electrochemical gradient favors H^+ movement from lumen to cell. Two mechanisms mediate active apical H^+ secretion. Approximately two-thirds occurs through the apical membrane Na^+-H^+ antiporter NHE3.³ This protein uses the inward Na^+ gradient to drive H^+ secretion. The Na^+-H^+ exchanger has a 1:1 stoichiometry and is electroneutral. In parallel with the Na^+-H^+ antiporter, there is an apical membrane H^+-ATPase that mediates approximately one-third of basal proximal tubular HCO_3^- absorption.

Both these H^+ transporters generate base in the cell, which must exit across the basolateral membrane to effect transepithelial transport. This primarily occurs through a basolateral $\text{Na}^+-\text{HCO}_3^-/\text{CO}_3^{2-}$ cotransporter.⁴ Because this protein transports the equivalent of two net negative charges, the negative cell voltage generated by the basolateral $\text{Na}^+/\text{K}^+-\text{ATPase}$ provides a strong favorable driving force for base efflux. The Na^+ carried on this transporter is moved out of the cell without requiring ATP. The $\text{Na}^+-3\text{HCO}_3^-$ cotransporter NBCe1, encoded by the gene *SLC4A4*, mediates most of the exit of base from the proximal tubule.⁵

Carbonic anhydrase II is present in the proximal tubular cell cytoplasm, and carbonic anhydrase IV is on the apical and basolateral membranes. Carbonic anhydrase (carbonate dehydratase) has a number of functions in the proximal tubule. Apical membrane carbonic anhydrase allows secreted H^+ ions to react with luminal HCO_3^- , forming H_2CO_3 , which rapidly dissociates to $\text{CO}_2 + \text{H}_2\text{O}$. This CO_2 diffuses across the apical plasma membrane into the cell. There the

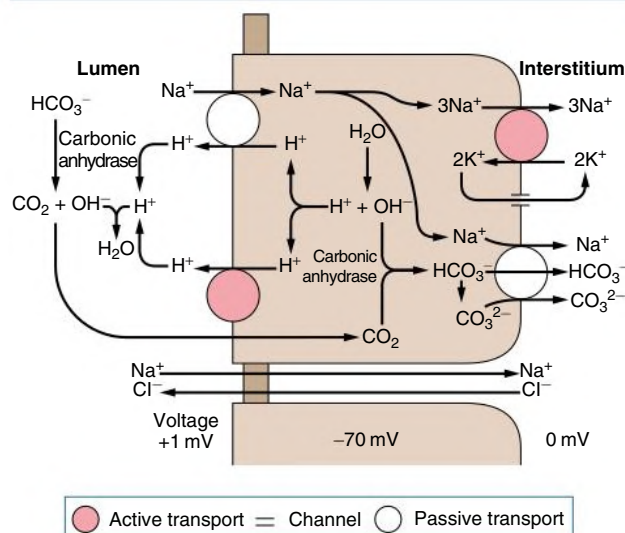
Proximal Tubule NaHCO_3 Reabsorption

Fig. 12.2 Proximal tubule Sodium Bicarbonate (NaHCO_3) Reabsorption. The secretion of H^+ into the proximal tubule lumen involves a $\text{Na}^+\text{-H}^+$ antiporter and an $\text{H}^+\text{-ATPase}$. Apical membrane H^+ secretion generates OH^- , which reacts with CO_2 to form HCO_3^- and CO_3^{2-} , and these exit with a Na^+ on the basolateral membrane $\text{Na}^+\text{-HCO}_3^- \text{-CO}_3^{2-}$ cotransporter. The Na^+ absorbed by the $\text{Na}^+\text{-H}^+$ antiporter exits the cell on the basolateral membrane $\text{Na}^+\text{-K}^+\text{-ATPase}$ and the $\text{Na}^+\text{-HCO}_3^- \text{-CO}_3^{2-}$ cotransporter. The K^+ that enters the cell on the $\text{Na}^+\text{-K}^+\text{-ATPase}$ exits on a basolateral membrane K^+ channel. Carbonic anhydrase catalyzes the conversion of HCO_3^- to CO_2 and OH^- in the lumen and the reverse reaction in the cell. Electrogenic H^+ secretion generates a small, lumen-positive voltage that generates a current flow across the paracellular pathway.

process is reversed, with use of cytoplasmic carbonic anhydrase, generating intracellular H^+ and HCO_3^- . This H^+ “replenishes” the H^+ secreted across the apical membrane, resulting in net movement of the HCO_3^- from the luminal solution to the cell cytoplasm. The intracellular HCO_3^- is then secreted across the basolateral plasma membrane, as described previously.

Thick Ascending Limb of the Loop of Henle

Tubular fluid arriving at the early distal tubule has a pH and serum $[\text{HCO}_3^-]$ similar to that in the late proximal tubule. Because there is significant water extraction in the loop of Henle, maintenance of a constant serum HCO_3^- concentration requires reabsorption of HCO_3^- . The majority of this HCO_3^- absorption occurs in the thick ascending limb (TAL) through mechanisms similar to those present in the proximal tubule (Fig. 12.3). The majority of apical membrane H^+ secretion is mediated by the $\text{Na}^+\text{-H}^+$ antiporter NHE3. As in the proximal tubule, the low intracellular Na^+ concentration maintained by the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$ provides the primary driving force for the antiporter. Base efflux across the basolateral membrane is mediated by a $\text{Cl}^- \text{-HCO}_3^-$ exchanger (AE2) and $\text{K}^+\text{-HCO}_3^-$ cotransport likely mediated by the $\text{K}^+\text{-Cl}^-$ cotransporter KCC4.⁶ These cells also possess an $\text{H}^+\text{-ATPase}$. The contribution of this pump to overall acidification in this segment is not clear.

Distal Nephron

Approximately 80% of the filtered HCO_3^- is reabsorbed in the proximal tubule; most but not all of the remainder is absorbed in the TAL. One function of the distal nephron is to reabsorb the remaining 5% of filtered HCO_3^- . In addition, the distal nephron must secrete a quantity of H^+ equal to that generated systemically by metabolism to maintain acid-base balance.

Hydrogen and Bicarbonate Ion Transport in the Thick Ascending Limb

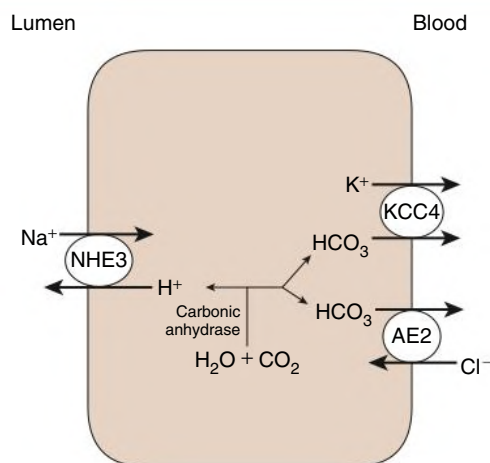


Fig. 12.3 Hydrogen (H^+) and Bicarbonate HCO_3^- Ion Transport in the Thick Ascending Limb. Apical H^+ secretion is mediated by a $\text{Na}^+\text{-H}^+$ antiporter. The low intracellular Na^+ concentration, maintained by the basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, provides the primary driving force for the antiporter. Both $\text{Cl}^- \text{-HCO}_3^-$ exchange and $\text{K}^+\text{-HCO}_3^-$ cotransport mediate base exit across the basolateral membrane.

The distal nephron is subdivided into several distinct portions that differ in their anatomy and acid secretory properties. Most of these segments transport H^+ and HCO_3^- into the luminal fluid, but the main segments appear to be in the collecting duct.⁷ The segments of the collecting duct include the cortical collecting duct (CCD), the outer medullary collecting duct, and the inner medullary collecting duct. There are two distinct cell types in the CCD that can be distinguished histologically: the principal cell and the intercalated (IC) cell. The principal cell reabsorbs Na^+ and secretes K^+ and is discussed later. Depending on chronic acid-base status, the CCD is capable of either H^+ or HCO_3^- secretion. These functions are mediated by two types of IC cells: the acid-secreting α -IC cell and the base-secreting β -IC cell. Both IC cell types are rich in carbonic anhydrase II.

Reabsorption of HCO_3^- in the distal nephron is mediated by apical H^+ secretion by the α -IC cell. Two transporters secrete H^+ : a vacuolar $\text{H}^+\text{-ATPase}$ and an $\text{H}^+\text{-K}^+\text{-ATPase}$ (Fig. 12.4). The vacuolar $\text{H}^+\text{-ATPase}$ is an electrogenic pump related to the H^+ pump present within lysosomes, the Golgi apparatus, and endosomes. The $\text{H}^+\text{-K}^+\text{-ATPase}$ uses the energy derived from adenosine triphosphate hydrolysis to secrete H^+ into the lumen and to reabsorb K^+ in an electroneutral fashion. The activity of the $\text{H}^+\text{-K}^+\text{-ATPase}$ increases in K^+ depletion and thus provides a mechanism by which K^+ depletion enhances both collecting duct H^+ secretion and K^+ absorption.⁸

Active H^+ secretion by the apical membrane generates an intracellular base that must exit the basolateral membrane. A basolateral $\text{Cl}^- \text{-HCO}_3^-$ exchanger (AE1) is the mechanism by which this base exit occurs. The Cl^- that enters the cell in exchange for HCO_3^- exits the cell through a basolateral membrane Cl^- conductance channel (see Fig. 12.4).

The HCO_3^- -secreting β -IC cell is a mirror image of the α -IC cell (Fig. 12.5). It possesses an $\text{H}^+\text{-ATPase}$ on the basolateral membrane, which mediates active H^+ extrusion. Alkali that is generated within the cell then exits on an apical membrane $\text{Cl}^- \text{-HCO}_3^-$ exchanger. This $\text{Cl}^- \text{-HCO}_3^-$ exchanger is distinct from the basolateral $\text{Cl}^- \text{-HCO}_3^-$ exchanger present in the α -IC cell and functions as an anion exchanger or Cl^- channel in the luminal membrane of epithelial cells.⁹ The SLC26A4 protein (pendrin) is a family member that mediates apical $\text{Cl}^- \text{-HCO}_3^-$ exchange in the β -IC cell of the kidney. The Na^+ -driven $\text{Cl}^- \text{-HCO}_3^-$

Secretion of H^+ in the α -Intercalated Cell of the Cortical Collecting Duct

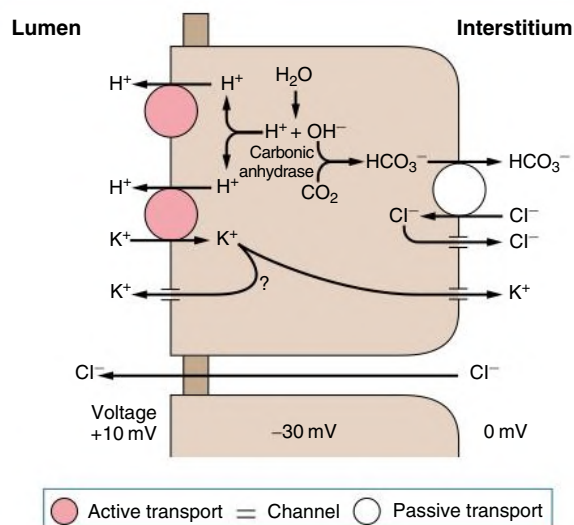


Fig. 12.4 Hydrogen (H^+) Secretion in the Cortical Collecting Duct α -Intercalated Cell. Secretion of hydrogen ions into the lumen by an H^+ -ATPase and an H^+ - K^+ -ATPase. Apical membrane H^+ secretion generates OH^- , which reacts with CO_2 to form HCO_3^- . This bicarbonate exits across the basolateral membrane on a Cl^- - HCO_3^- exchanger, a member of the anion exchanger-1 (AE1) family and a truncated form of the red blood cell AE1 Cl^- - HCO_3^- exchanger. The Cl^- that enters the cell on the exchanger recycles across a basolateral membrane Cl^- channel. The K^+ that enters the cell on the H^+ - K^+ -ATPase appears to be able either to recycle across the apical membrane or exit across the basolateral membrane, depending on the potassium balance of the individual. Carbonic anhydrase catalyzes the conversion of CO_2 and OH^- to HCO_3^- in the cell. Electrogenic H^+ secretion generates a lumen-positive voltage that generates a current flow across the paracellular pathway.

exchanger (NDCBE) colocalizes with pendrin on the apical membrane and together may explain a component of electroneutral NaCl reabsorption in the collecting duct that is thiazide sensitive.¹⁰

The other cortical collecting tubule cell type is the principal cell, which also regulates acid-base transport, although indirectly. Principal cells mediate electrogenic Na^+ reabsorption that results in a net negative luminal charge (Fig. 12.6). The greater this negative charge, the lesser is the electrochemical gradient for electrogenic proton secretion and therefore the greater the rate of net proton secretion. Thus, factors that stimulate Na^+ reabsorption indirectly regulate the H^+ secretory rate.

The medullary collecting duct possesses mechanisms only for H^+ secretion. This H^+ secretion is mediated by α -IC cells but also by cells that appear morphologically distinct from IC cells but are functionally similar.

Net Acid Excretion

For the kidney to generate NAE, it must both reabsorb filtered HCO_3^- and excrete titratable acids and ammonia. Several weak acids, such as phosphate, creatinine, and uric acid, are filtered at the glomerulus and can buffer secreted protons. Of these, phosphate is the most important because of its favorable pK_a of 6.80 and its relatively high rate of urinary excretion (~ 25 – 30 mmol/day). However, the capacity of phosphate to buffer protons is maximized at a urine pH of 5.8, and acid-base disturbances generally do not induce substantial changes in urinary phosphate excretion. Other titratable acids, such as creatinine and uric acid, are limited by their lower excretion rate, which is not

Bicarbonate Secretion by Cortical Collecting Duct β -Intercalated Cell

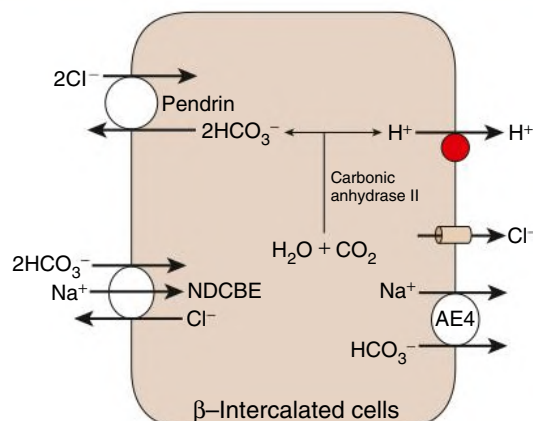


Fig. 12.5 Bicarbonate Secretion by the Cortical Collecting Duct β -Intercalated Cell. H^+ is secreted into the interstitium by an H^+ -ATPase. The OH^- generated by basolateral membrane H^+ secretion reacts with CO_2 to form HCO_3^- , which exits across the apical membrane on a Cl^- - HCO_3^- exchanger (pendrin). The Cl^- that enters the cell on the exchanger exits across a basolateral membrane Cl^- channel. Carbonic anhydrase II catalyzes the conversion of CO_2 and OH^- to HCO_3^- in the cell. The Na^+ -driven Cl^- - HCO_3^- exchanger (NDCBE) colocalizes with pendrin on the apical membrane and mediates thiazide-sensitive electroneutral NaCl reabsorption in this segment.

Transport of Na^+ in the Principal Cell of the Cortical Collecting Duct

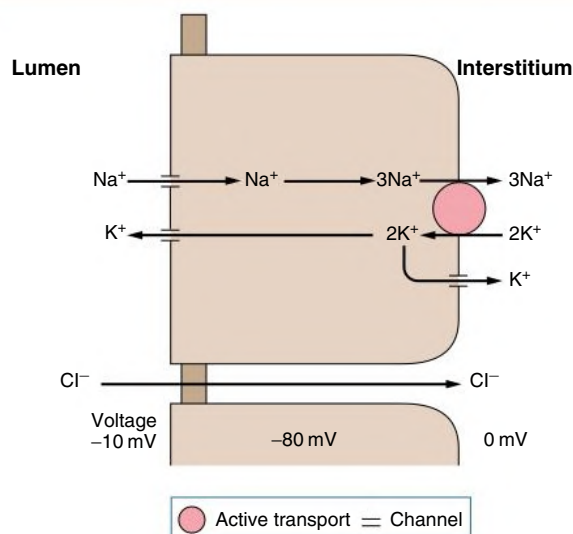


Fig. 12.6 Sodium Transport in the Cortical Collecting Duct Principal Cell. Electrogenic Na^+ absorption is mediated by the Na^+ channel. The Na^+ enters the cell across the apical membrane channel and exits the cell on the basolateral membrane Na^+ , K^+ -ATPase. The K^+ that enters the cell on the basolateral Na^+ , K^+ -ATPase can be secreted into the luminal fluid by an apical membrane K^+ channel. Electrogenic Na^+ absorption establishes a lumen-negative voltage that drives a paracellular current.

dramatically changed in response to acid-base disturbances. Titratable acid excretion is a minor component of the increase in NAE in response to metabolic acidosis (Fig. 12.7).

Changes in Net Acid Excretion in Response to Chronic Metabolic Acidosis

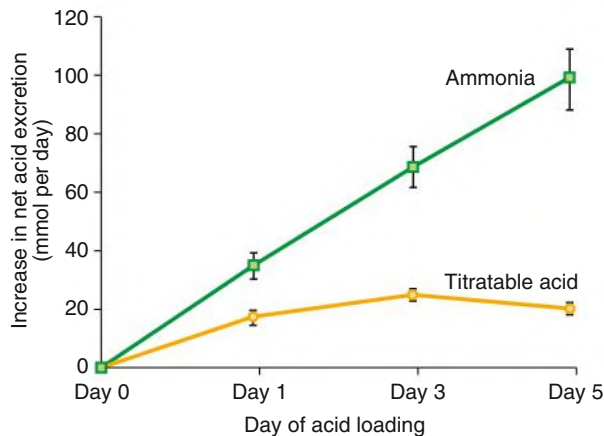


Fig. 12.7 Changes in Net Acid Excretion. Chronic metabolic acidosis increases net acid excretion dramatically over several days; shown quantitatively are the increases in the two major components of net acid excretion: titratable acids and ammonia. Titratable acid excretion increases slightly and predominantly in the first 24 to 48 hours. In contrast, urinary ammonia excretion progressively increases over 7 days and is responsible for most of the increase in net acid excretion in chronic metabolic acidosis. (Data from Elkinton JR, Huth EJ, Webster GD Jr, McCance RA. The renal excretion of hydrogen ion in renal tubular acidosis. I. Quantitative assessment of the response to ammonium chloride as an acid load. *Am J Med.* 1960;36:554–575.)

Ammonia Metabolism

Quantitatively, the most important component of NAE is the $\text{NH}_3/\text{NH}_4^+$ system.¹² Unlike for titratable acids, the rate of ammonia (NH_3) production and excretion varies according to physiologic needs. Under normal circumstances, ammonia excretion accounts for approximately 60% of total NAE, and in chronic metabolic acidosis, almost the entire increase in NAE is caused by increased NH_3 metabolism. Ammonia metabolism involves interplay among the proximal tubule, TAL, and collecting duct.

The proximal tubule is responsible for both ammonia production and luminal secretion. Ammonia is synthesized in the proximal tubule predominantly from glutamine metabolism through enzymatic processes in which phosphoenolpyruvate carboxykinase and phosphate-dependent glutaminase are the rate-limiting steps. This results in production of two ammonium (NH_4^+) and two HCO_3^- ions from each glutamine ion. Ammonia is then preferentially secreted into the lumen. The primary mechanism for this luminal secretion appears to be NH_4^+ transport by the apical Na^+-H^+ antiporter NHE3¹³ (Fig. 12.8).

Metabolic acidosis increases the mobilization of glutamine from skeletal muscle and intestinal cells. Glutamine is preferentially taken up by the proximal tubular cell through the Na^+ - and H^+ -dependent glutamine transporter SNAT3. This transporter is a member of the SCL38 gene family of Na^+ -coupled neutral amino acid transporters. SNAT3 expression increases several-fold in metabolic acidosis, and it is preferentially expressed on the cell's basolateral surface, where it is poised for glutamine uptake.¹⁴ The increase in plasma cortisol that typically accompanies metabolic acidosis plays a role in this transporter's upregulation.¹⁵ Metabolic acidosis also causes increased expression and activity of phosphate-activated glutaminase and glutamate dehydrogenase.

Most of the ammonia that leaves the proximal tubule does not reach the distal tubule. Thus, there is transport of ammonia out of the

loop of Henle. This ammonia transport appears to occur predominantly in the TAL and is mediated by at least three mechanisms¹⁶ (Fig. 12.9). First, the lumen-positive voltage provides a driving force for passive paracellular NH_4^+ transport out of the TAL. Second, NH_4^+ can be transported out of the lumen by the furosemide-sensitive $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ transporter. Third, NH_4^+ can leave the lumen across the apical membrane K^+ channel of the TAL cell. NH_4^+ exits the cell via the Na^+-H^+ exchanger NHE4, functioning in $\text{Na}^+-\text{NH}_4^+$ mode.

In addition, ammonia is secreted by the collecting duct. Although the traditional thought was that $\text{NH}_3/\text{NH}_4^+$ then enters the collecting duct by nonionic diffusion driven by the acid luminal pH, increasing evidence suggests that the nonerythroid glycoproteins Rhbg and Rhcg may be involved in collecting duct ammonia secretion.^{17,18}

Ammonia excretion can be regulated by three mechanisms. First, ammonia synthesis in the proximal tubule can be regulated. Chronic acidosis and hypokalemia increase ammonia synthesis, whereas hyperkalemia suppresses ammonia synthesis. Second, ammonia delivery from the proximal tubule to the medullary interstitium can be regulated. In particular, chronic metabolic acidosis increases expression of both NHE3 and the loop of Henle $\text{Na}^+-\text{K}^+-2\text{Cl}^-$ cotransporter. Hyperkalemia can inhibit NH_4^+ reabsorption from the TAL. The combined effects of decreased NH_3 synthesis in the proximal tubule and interference in NH_4^+ reabsorption in the thick limb may explain the low urinary $[\text{NH}_4^+]$ found in hyperkalemic distal renal tubular acidosis. The reduced availability of ammonia to serve as a urinary buffer leads to a reduction in distal H^+ secretion and development of metabolic acidosis. Also, any interstitial renal disease that destroys renal medullary anatomy may decrease medullary interstitial $[\text{NH}_3/\text{NH}_4^+]$ transfer. Third, mechanisms that regulate collecting duct H^+ secretion or ammonia transporter expression can regulate ammonia entry into the collecting duct and ammonia excretion. Importantly, the primary mechanisms require synthesis of new proteins to increase both ammonia production and transport. Accordingly, changes in ammonia excretion may be delayed, and the maximal renal response to chronic metabolic acidosis requires 4 to 7 days.

REGULATION OF KIDNEY ACIDIFICATION

The regulation of acid-base balance requires an integrated system that precisely regulates proximal tubular $\text{H}^+-\text{HCO}_3^-$ transport, distal nephron $\text{H}^+-\text{HCO}_3^-$ transport, and ammonia synthesis and transport.

Blood pH

The regulation of acid-base balance requires that net H^+ excretion increase in states of acidosis and decrease in states of alkalosis. This form of regulation involves both acute and chronic mechanisms. In the proximal tubule, acute decreases in blood pH increase the rate of HCO_3^- absorption, and acute increases in blood pH inhibit HCO_3^- absorption. These alterations in the rate of HCO_3^- absorption occur whether the change in pH is the result of changes in Paco_2 or serum $[\text{HCO}_3^-]$. Similarly, in the collecting duct, acute changes in peritubular serum $[\text{HCO}_3^-]$ and pH regulate the rate of H^+ secretion.

In addition to acute regulation, mechanisms exist for chronic regulation. Chronic acidosis or alkalosis leads to parallel changes in the activities of the proximal tubule apical membrane Na^+-H^+ antiporter and basolateral membrane $\text{Na}^+-\text{HCO}_3^-/\text{CO}_3^{2-}$ cotransporter. Metabolic acidosis acutely increases the kinetic activity of NHE3 through direct pH effects and by phosphorylation; chronic acidosis increases the number of NHE3 transporters.^{19,20} In addition, chronic acidosis increases proximal tubular ammonia synthesis by increasing the activities of the enzymes involved in ammonia metabolism.

Ammonia Synthesis and Transport in the Proximal Tubule

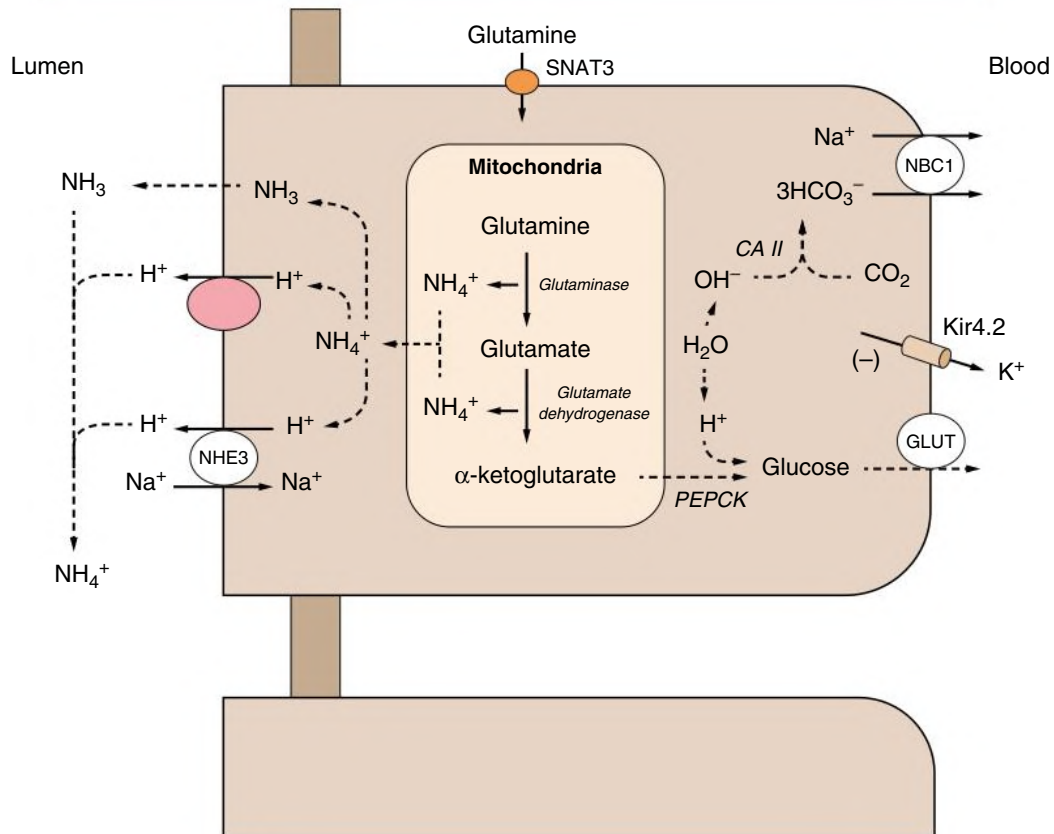


Fig. 12.8 Ammonia Synthesis and Transport in the Proximal Tubule. Metabolic acidosis and hypokalemia stimulate proximal ammonia synthesis by stimulating the uptake of glutamine through SNAT3 and increasing the activities of the enzymes involved in ammonia metabolism. The generation of ammonia is the result of glutamine metabolism by enzymes closely linked to gluconeogenesis. Hyperkalemia suppresses ammoniagenesis by increasing cell pH. Increased extracellular K^+ concentration leads to depolarization of the proximal tubular cell (decrease in electronegativity of the cell) as sensed through the inwardly rectifying K^+ subchannel Kir4.2 encoded by *Kcnj15* located on the basolateral membrane. This change results in less HCO_3^- exit coupled to Na^+ via NBCe1 because the cotransporter carries a net negative charge and is driven by intracellular electronegativity. The net effect is increased cell pH and downregulation of proteins involved in ammoniagenesis.

Ammonia Transport in the Thick Ascending Limb of Henle's Loop

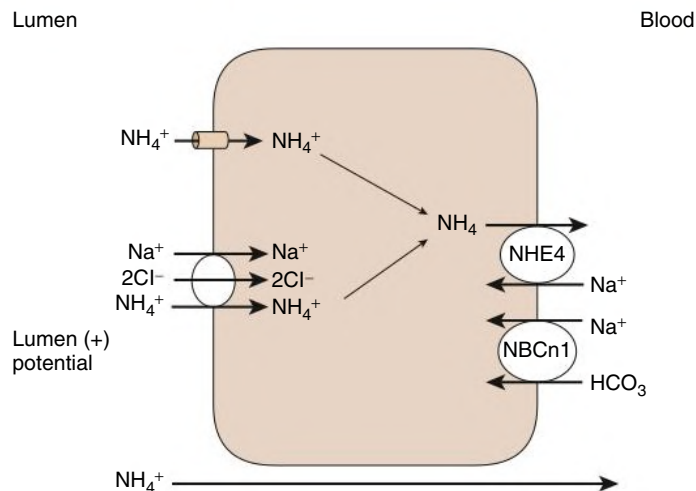


Fig. 12.9 Ammonia Transport in the Thick Ascending Limb of Henle's Loop. In addition to reabsorption through the paracellular pathway driven by the lumen-positive potential, ammonium can substitute for K^+ on the $Na^+-K^+-2Cl^-$ transporter and the apical membrane K^+ channel. NH_4^+ exits the cell via the Na^+-H^+ exchanger NHE4. The basolateral $Na^+-HCO_3^-$ cotransporter (NBCn1) functions to maintain an optimal cell pH to facilitate the large transcellular ammonium flux.

The CCD is also modified by chronic acid-base changes. Long-term increases in dietary acid lead to an increase in H^+ secretion, whereas long-term increases in dietary alkali lead to an increased capacity for HCO_3^- secretion.²¹ This effect is mediated by changes in the relative number of α - and β -IC cells. For example, during metabolic acidosis, the number of α -IC cells increases and the number of β -IC cells decreases, without a change in the total number of IC cells. Recent evidence suggests that the extracellular protein hensenin may be involved in the switch between the predominant IC cell types.²²

Mineralocorticoids, Distal Sodium Delivery, and Extracellular Fluid Volume

Mineralocorticoid hormones are key regulators of distal nephron and collecting duct H^+ secretion. Two mechanisms appear to be involved. First, mineralocorticoid hormone stimulates Na^+ absorption in principal cells of the CCD (see Fig. 12.6). This leads to a more lumen-negative voltage that then stimulates H^+ secretion. This mechanism is indirect in that it requires the presence of Na^+ and of Na^+ transport. The second mechanism is the direct activation of H^+ secretion by mineralocorticoids. This effect is chronic, requiring long exposure, and involves parallel increases in apical membrane H^+ -ATPase and basolateral membrane Cl^- - HCO_3^- exchanger activity.

Plasma Volume

Changes in plasma volume have important effects on acid-base homeostasis. This effect appears to be related to a number of factors. First, volume contraction is associated with a decreased GFR, which lowers the filtered load of HCO_3^- and decreases the load placed on the tubules to maintain NAE. Volume contraction also acutely decreases the paracellular permeability of the proximal tubule. This will decrease HCO_3^- back-leak around cells, thereby increasing net bicarbonate reabsorption by the proximal tubule. Third, chronic volume contraction is associated with an adaptive increase in the activity of the proximal tubule apical membrane

Na^+ - H^+ antiporter NHE3. Because this transporter contributes to both $NaHCO_3$ and $NaCl$ absorption, both these capacities will be increased with chronic volume contraction. Further, volume contraction limits distal delivery of chloride. In the presence of chronic metabolic alkalosis, the CCD is poised for HCO_3^- secretion. However, collecting duct HCO_3^- secretion requires luminal Cl^- and is inhibited by Cl^- deficiency.

Potassium

Potassium deficiency is associated with an increase in renal NAE. This effect is multifactorial. First, chronic K^+ deficiency increases the proximal tubule apical membrane Na^+ - H^+ antiporter and basolateral membrane Na^+ - HCO_3^- - CO_3^{2-} cotransporter activities. This effect is similar to that seen with chronic acidosis and may be caused by intracellular acidosis. Chronic K^+ deficiency also increases proximal tubular ammonia production. Finally, chronic K^+ deficiency leads to an increase in collecting duct H^+ secretion. This appears to be related to increased activity of the apical membrane H^+ - K^+ -ATPase. Such an effect increases the rate of H^+ secretion and the rate of K^+ reabsorption in the collecting duct. In addition, ammonia, whose production is stimulated by hypokalemia, has direct effects that stimulate collecting duct H^+ secretion. Counterbalancing these effects is that K^+ deficiency decreases aldosterone secretion, which can inhibit distal acidification. Thus, in normal individuals, the net effect of K^+ deficiency is typically a minor change in acid-base balance. However, in patients with non-suppressible mineralocorticoid secretion (e.g., hyperaldosteronism, Cushing syndrome), K^+ deficiency can greatly stimulate renal acidification and cause profound metabolic alkalosis.

Hyperkalemia appears to have opposite effects on renal acidification. The most notable effect of hyperkalemia is inhibition of ammonia synthesis in the proximal tubule and ammonia absorption in the loop of Henle, resulting in inappropriately low levels of urinary ammonia excretion. This contributes to the metabolic acidosis seen in patients with hyperkalemic distal (type 4) renal tubular acidosis.

SELF-ASSESSMENT QUESTIONS

- Which of the following is *true* regarding the effect of chronic K^+ deficiency on renal acidification?
 - Decreases basolateral membrane Na^+ - HCO_3^- - CO_3^{2-} cotransporter activity
 - Increases proximal tubular ammonia production
 - Decreases collecting duct H^+ secretion
 - Increases aldosterone secretion
- Which one of the following is *true* regarding ammonia handling in the kidney?
 - Ammonia synthesized in the proximal tubule is primarily reabsorbed in the cortical collecting duct.
 - Ammonia is reabsorbed passively in the thick ascending limb driven by the lumen-negative voltage.
 - NH_4^+ is transported into the lumen of the thick ascending limb by the Na^+ - K^+ - Cl^- transporter.
 - Nonthyroid glycoproteins Rhbg and Rhcg are involved in collecting duct ammonia secretion.
- Which of the following is *true* regarding net acid production by the kidney?
 - Sulfur-containing amino acids are metabolized to CO_2 .
 - Metabolism of lysine and arginine yield base on metabolism.
 - Animal foods are high in protein and organophosphates and provide a net acid diet.
 - Daily volatile acid production is approximately 1 mmol H^+ .

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